Analytical Chemistry Cumulative Exam (Electroanalytical Chemistry)
November 7th, 2025

Instructions

Please clearly show all your workings. Partial credit will be awarded for demonstrating relevant knowledge and understanding. A correctly set up problem containing minor calculation errors could still pass this cume! As always, please use the appropriate significant figures and units.

Question

Provide **two quantitative, clearly labelled sketches** showing the results of performing cyclic voltammetry of the following system.

A quiescent aqueous solution of 2 mM $Ru(NH_3)_6Cl_3$ containing 0.5 M KCl at room temperature. Show a single voltammetric cycle starting at +0.15 V scanning in a negative direction, and reversing at -0.3 V back to +0.15 V (all vs an SCE reference electrode) with a scan rate of 10 mV/s.

i) At a 1-mm radius Pt disk electrode
ii) At a 25-µm diameter Pt microdisk electrode

A 'quantitative sketch' means that you should use numbers to indicate key values of currents/potentials within your plots. You should calculate these values independently before making your sketch. Show your working and label your axes with appropriate units.

You may find some of the following information and the equations on the final page of this exam to be useful in answering this question. Note: You will not need everything that is provided.

$E^0(Ru(NH_3)_6^{3+}/Ru(NH_3)_6^{2+})$ = -45 mV vs SCE; the diffusion coefficients of:

$Ru(NH_3)_6^{3+} = 8.43\times10^{-6}\ cm^2/s$ and $Ru(NH_3)_6^{2+} = 1.19 \times 10^{-5}\ cm^2/s$.

You may assume that the measurements were carefully performed by a competent experimenter who didn't encounter any instrumental problems. Before measuring, the experimenter purged all O_2 from the solution by bubbling with Ar. The $Ru(NH_3)_6^{3+}/Ru(NH_3)_6^{2+}$ redox couple shows very fast electron transfer kinetics, and so you may assume it is reversible.

Information has been provided to you in a range of commonly used units. Note, these are not all base SI units – please be careful to account for this in your answers.

$$F = 96485 \text{ C/mol}$$

$$R = 8.3144 \text{ J/(mol K)}$$

$$E = E^0 + \frac{RT}{nF} \ln \frac{a_{\text{ox}}}{a_{\text{red}}}$$

$$E = E^{0\prime} + \frac{RT}{nF} \ln \frac{[\text{ox}]}{[\text{red}]}$$

$$E = E^0 + \frac{RT}{nF} \ln \frac{\prod_j^{\text{O side}} a_j^{\nu_j}}{\prod_j^{\text{R side}} a_j^{\nu_j}}$$

$$\Delta G = -nFE$$

$$E_{\text{rxn}} = E_{\text{cell}} = E_{\text{red}} - E_{\text{ox}}$$

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$i_{\text{p}} = 2.69 \times 10^5 n^{\frac{3}{2}} A D_{\text{O}}^{\frac{1}{2}} C_{\text{O}}^* \nu^{\frac{1}{2}}$$

$$i(t) = \frac{nFAD_O^{\frac{1}{2}} C_{\text{o}}^*}{\pi^{\frac{1}{2}} t^{\frac{1}{2}}}$$

$$i_{\text{lim,disk}} = 4naFDC^*$$

$$k_{\text{f}} = k^0 \exp[-\alpha f\left(E - E^{0\prime}\right)]$$

$$k_{\text{b}} = k^0 \exp[(1-\alpha) f\left(E - E^{0\prime}\right)]$$

where $f = F/RT$